

The Utility Problem

At any time, a wiring error may occur when an electricity meter is installed. This error (among others) can result in incorrect measurements and, therefore, incorrect customer billing. The exchange between consumption and generation registers might occur for customers with an installed power source (e.g. a photovoltaic panel). Because of this, the customer would then be incorrectly billed, and the utility could receive less revenue from that customer.

Problems also arise in grid calculations because erroneous data enter into these calculations. If such an error were to accumulate in an area, it could fundamentally affect further analysis and, therefore, future grid plans in that area. Negative consequences may include safety, reliability and other issues.

Utilities connect a large number of customers on a daily basis, and hence this problem can occur and grow. Therefore, it is important to have methods to reveal such issues and gain control over these occurrences.

The Data Solutions and Methods

To solve this task, time-series meter data from all available registers are needed. With the Awesense Energy Transition Platform, this data is very easy to obtain. The predefined functions for obtaining aggregated data make this task very easy - to avoid having to work with a lot of data, only hourly meter data is required for the below-described method.

In this scenario, the time-series data is checked for customer sites with a photovoltaic (PV) source installed. Meter registers are checked, and it is checked if there is no solar power generation during the nighttime because it is impossible to have solar power generation during the nighttime. If there is a generation at night, a flag can be created, or a notification will be received about that meter location - including any additional information, such as an address, GPS coordinates, etc. The results can also be displayed on a map so that we are able to prioritize repairs if there are multiple incorrect connections in an area.

Data required for this use case:

- Time-series meter data:
 - Synchronized in time and assigned geospatially to corresponding consumption points - meters,
- Additional information related to the consumption point - address, GPS coordinates, etc.

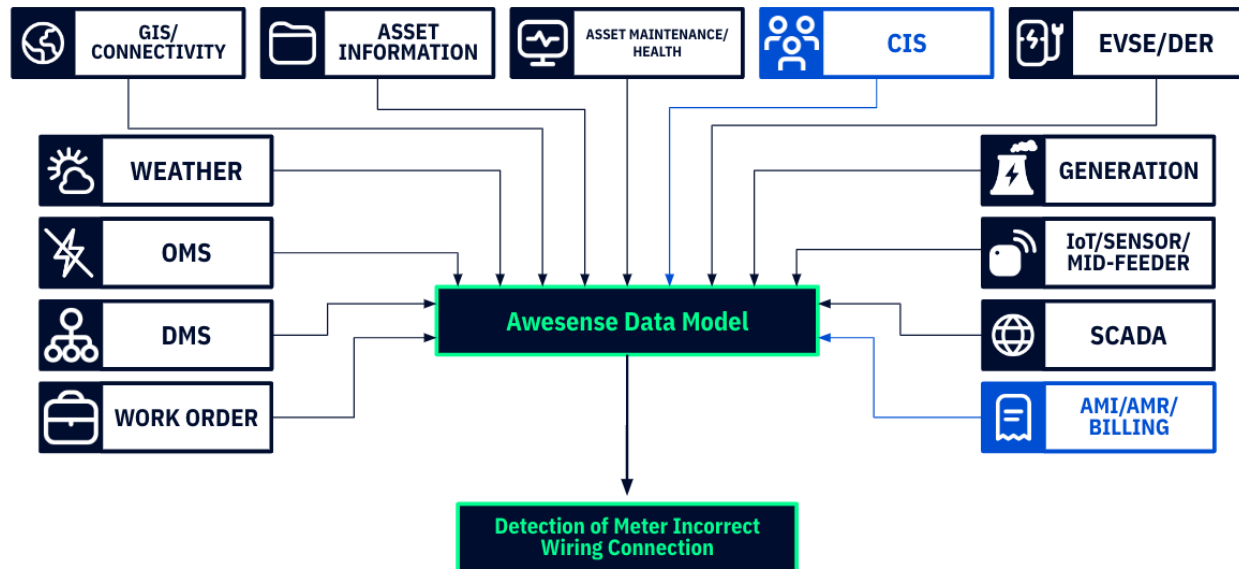
Detection of Meter Incorrect Wiring Connection

[UC37-01] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. A GitHub link to the Jupyter notebook can be provided upon request.